PRASAD V. POTLURI SIDDHARTHA INSTITUTE OF TECHNOLOGY

(Autonomous Institution - Affiliated to JNTUK | Accredited by NAAC A+)

Department of Computer Science & Engineering

Full Stack Development (FSD) Capstone Project Report | Academic Year 2026-27

StudySwap

?Find. Share. Learn. Together.?

A unified academic resource sharing platform connecting students to discover, rate, review, and download branch-specific verified lecture notes and question banks.

LIVE PROJECT DEPLOYMENT & REPOSITORY URLS

? Live Web Application (Vercel):	`https://study-swap-find-share-learn-togethe.vercel.app`
? GitHub Repository:	`https://github.com/Shyam062006/StudySwap-Find.Share.Learn.Together`
? Production REST API (Render):	`https://studyswap-api.onrender.com/api/health`
? Cloud Database (MongoDB Atlas):	cluster0.d35bywl.mongodb.net/studyswap (24/7 Multi-Region Cluster)

PROJECT TEAM MEMBERS (DEPARTMENT OF CSE)

S.No	Student Name	Roll Number	Project Role & Contribution	GitHub Profile
1	**Shyam Sundar Penti**	`24501A05H9`	Full-Stack Lead & System Architect	`@Shyam062006`
2	**Neelish Sri Charan**	`24501A05H8`	Frontend & UI/UX Specialist	`@neelsihsricharan`
3	**Peddasingu Praveen Kumar**	`24501A05H5`	Backend & Database Engineer	`@Peddasing-Praveen-kumar`
4	**Sai Subhash**	`24501A05I2`	Quality Assurance & API Testing	`@SaiSubhash-07`

EVALUATION DEMO ACCOUNTS (BRANCH: CSE | SEMESTER: 5)

??? Student 1: Praveen (Uploader)

```
Email: praveen@studyswap.dev
Password: Student123!
```

??? Student 2: Neelish (Reviewer & Downloader)

```
Email: neelish@studyswap.dev
Password: Student123!
```

TECHNICAL SPECIFICATIONS & ARCHITECTURE OVERVIEW

Comprehensive breakdown of full-stack modules, design patterns, and engineering decisions.

Frontend Presentation Tier

? React 18 SPA with Vite & React Router v6
? Red Noir Design System, Dark Mode Default, Lucide Icons
? Responsive Grids (Desktop 3-col, Tablet 2-col, Mobile 1-col)

Backend Application Tier

? Node.js & Express.js RESTful API architecture
? Strict input validation with express-validator
? Helmet security headers, CORS origin filtering, Rate Limiters

Database & Storage Tier

? MongoDB Atlas Cloud with official native driver (no ORM)
? Indexed text search on title, description, subject, and tags
? Multer storage engine with MIME validation (PDF, DOCX, PPTX)

Authentication & Security

? JWT tokens stored in httpOnly, Secure, SameSite cookies
? Bcrypt salt rounds (10-12) for secure password hashing
? Strict prevention of self-rating and duplicate reviews

CORE MODULES IMPLEMENTED & VERIFIED

1. Central Material Hub

Unified repository where all university materials reside. Multi-attribute smart filtering by Branch (CSE, ECE, MECH, etc.), S

2. Instant Upload & Auto-Approval

Pure student-to-student sharing platform. Uploaded notes, records, and question banks are automatically validated and publish

3. Peer Rating & Interactive Reviews

Allows fellow students to rate study materials (1-5 stars) and write detailed qualitative reviews. Includes self-review preve

4. Gamified Student Leaderboard

Live ranking table tracking student contributions in real-time. Students earn contribution points for approved uploads, recei

5. Bookmarks & Personal Download Hub

Allows students to save and bookmark important materials for exam preparation. Full tracking of personal uploads, download hi

PROJECT VERIFICATION & STATUS: 100% OPERATIONAL

? Complete end-to-end integration verified: Registration -> Login -> Material Upload -> Hub Search -> Review -> Leaderboard.
? Verified live in cloud: Frontend hosted on Vercel, Backend API on Render, Database on MongoDB Atlas.
? All automated tests passed. Ready for university evaluation and laboratory viva-voce.